Ali Diba

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Summary

15+ years of experience in ***computer vision and machine learning*** and solving different problems in the field of video and image understanding and deep neural networks. Passionate about designing scalable deep-tech solutions at scale. My major experience is creating and leading cross-functional R&D teams (20+ people) to deliver high-quality AI-first large-scale products, both as a scientist and leader.

Principal Interests

Computer vision, Video understanding, AI-hardware CV models, Foundational Models, Large-scale visual understanding, Deep generative models, Object detection & Segmentation, Large language models, and Multi-modal models.

Professional Experience

Principal Investigator & Scientist — July 2025 - Present
QCRI, Doha
- Leading research on novel large-scale multi-modal models for video and image understanding.
- Novel multi-modal large-scale models for social computing and humanitarian AI
- Multi-modal medical and cancer monitoring models
- Supervising research scientists and engineers.

Staff Research Scientist and Research Lead — Sep 2023 - July 2025
Lunit Inc., Amsterdam
- Leading research and development of novel solutions for cancer screening in medical images and devices.
- Supervising research scientists and engineers.
- Projects: Large vision-language models, multi-modal deep learning models

Senior Researcher — Oct 2022 - Sep 2023
PSI, Leuven
- Leading research on video understanding and self-supervised learning
- Supervising Ph.D. projects with Prof. Luc Van Gool
- Organizing international workshops and tutorials on computer vision
- Projects: Large vision models, multi-modal learning models, Vision-Language models

Chief Technology Officer/ Co-Founder — 2021 - June 2023
BuyBuy BV (Stealth Startup), Leuven, Belgium
- CTO and Principal technical lead
- Design and implementing efficient **CV/ML on AI-edge** hardware.

Chief Technology Officer/ Co-Founder — 2015 - 2020
Sensifai BV (KU Leuven Spin-off), Leuven, Belgium
- Principal AI and computer vision lead

- Leading design and implementing computer vision SaaS solutions on AWS and Huawei platforms
- AI models R&D on **Huawei**, **Qualcomm**, and **Nvidia** AI/Edge hardware
- Managing the launch partnership of AWS SageMaker
- Cloud/Edge computer vision models, Large-scale video and image tagging, Facial recognition, Crowd analysis in video streams

Education

Ph.D. Electrical Engineering, Computer Vision 2021
KU Leuven, Leuven, Belgium

- Ph.D. research in computer vision under the supervision of Prof. Luc Van Gool. Dissertation title: Large-scale video understanding
- Nominated for ELLIS PhD Award 2022

M.Sc. Computer Engineering 2013
Sharif University of Technology, Tehran, Iran

- Focus areas: computer vision and scene understanding (thesis: Scene classification By Segmented frames from video streams) under the supervision of Prof. Mohammad Ghanbari

Key Publications

See also my google scholar page. (**citations>3200 and H-index 24**)
The listed publications are peer-reviewed conferences or journal articles and top-tier in the field of computer vision. CVPR, NeurIPs, ICCV, ECCV, and MICCAI are highly competitive, with acceptance rates of less than 25%, and among the most impactful conferences in computer science. CVPR is the most cited IEEE conference with the highest impact in Engineering and Computer Science.

28. F.Zheng, A.Diba, et al, **"Breast Cancer VLMs: Clinically Practical Vision-Language Train-Inference Models"**, **CVAMD, ICCV 2025**
27. H. Ardi, A. Jahanshahi, A. Diba , **"T-SiamTPN: Temporal Siamese Transformer Pyramid Networks for Robust and Efficient UAV Tracking"**, **Arxiv 2025**
26. F.Zheng, A.Diba, et al, **"MV-MLM: Bridging Multi-View Mammography and Language for Breast Cancer Diagnosis and Risk Prediction"**, **ICCV 2025**
25. L.Dillard, et al, **"SelectiveKD: A semi-supervised framework for cancer detection in DBT through Knowledge Distillation and Pseudo-labeling"**, **MICCAI, 2024**
24. M.Heidari, et al, **"Computation-Efficient Era: A Comprehensive Survey of State Space Models in Medical Image Analysis"**, **Arxiv, 2024**
23. Ali Diba, Vivek Sharma, Luc Van Gool, **"Spatio-Temporal Convolution-Attention Video Network"**, **ICCV, 2023**
22. Ali Diba, et. al, **"4th Large scale Holistic Video Understanding"**, **CVPR, 2023**
21. Ali Diba, Luc Van Gool, **"Long-Term Video Understanding and Reasoning"**, Journal Under review, 2022
20. Mohsen Fayyaz, Emad Bahrami Rad, Ali Diba, Mehdi Noroozi, Ehsan Adeli, Luc Van Gool, Juergen Gall, **"3D CNNs with Adaptive Temporal Feature Resolutions"**, ***CVPR*, 2021**

19. M Saquib Sarfraz, Naila Murray, Vivek Sharma, Ali Diba, Luc Van Gool, Rainer Stiefelhagen, **"Temporally-Weighted Hierarchical Clustering for Unsupervised Action Segmentation"**, ***CVPR*, 2021**
18. Ali Diba, Vivek Sharma, Reza Safdari, Dariush Lotfi, Saquib Sarfraz, Rainer Stiefelhagen, Luc Van Gool, **"Vi2CLR: Video and Image for Visual Contrastive Learning of Representation"**, ***ICCV*, 2021**
17. Callemein Timothy, Tom Roussel, Ali Diba, Boes Wim, Tuytelaars Tinne, Goedemé Toon, **"Show me where the action is!"**, *Journal of Multimedia Tools and Applications*, 2021
16. Ali Varamesh, Ali Diba, Tinne Tuytelaars, Luc Van Gool, **"Self-Supervised Ranking for Representation Learning"**, ***Neurips*, 2020**
15. Ali Diba, Mohsen Fayyaz, Vivek Sharma, Manohar Paluri, Jürgen Gall, Rainer Stiefelhagen, Luc Van Gool, **"Large scale holistic video understanding"**, ***ECCV*, 2020**
14. Ali Diba, Vivek Sharma, Luc Van Gool, Rainer Stiefelhagen, **"DynamoNet: Dynamic Action and Motion Network"**, ***ICCV*, 2019**
13. Ali Diba, Vivek Sharma, Rainer Stiefelhagen, Luc Van Gool, **"Weakly Supervised Object Discovery by Generative Adversarial & Ranking Networks"**, ***CVPR-CEFRL*, 2019**
12. Ali Diba, Mohsen Fayyaz, Vivek Sharma, M Mahdi Arzani, Rahman Yousefzadeh, Juergen Gall, Luc Van Gool, **"Spatio-Temporal Channel Correlation Networks for Action Classification"**, ***ECCV*, 2018**
11. Ali Diba, Mohsen Fayyaz, Vivek Sharma, A Hossein Karami, M Mahdi Arzani, Rahman Yousefzadeh, Luc Van Gool, **"Temporal 3D ConvNets using Temporal Transition Layer"**, ***CVPR*, 2019**
10. Dries Hulens, Bram Aerts, Punarjay Chakravarty, Ali Diba, Toon Goedemé, Tom Roussel, Jeroen Zegers, Tinne Tuytelaars, Luc Van Eycken, Luc Van Gool, Hugo Van Hamme, Joost Vennekens, **"The cametron lecture recording system: High-quality video recording and editing with minimal human supervision"**, ***ICMM*, 2018**
9. Vivek Sharma, Ali Diba, Davy Neven, Michael S Brown, Luc Van Gool, Rainer Stiefelhagen, **"Classification Driven Dynamic Image Enhancement"**, ***CVPR*, 2018**
8. Ali Diba, Vivek Sharma, Luc Van Gool, **"Deep Temporal Linear Encoding Networks"**, ***CVPR*, 2017**
7. Ali Diba, Vivek Sharma, Ali Pazandeh, Hamed Pirsiavash, Luc Van Gool, **"Weakly Supervised Cascaded Convolutional Networks"**, ***CVPR*, 2017**
6. Ali Diba, Ali Mohammad Pazandeh, Luc Van Gool, **"Deep visual words: Improved fisher vector for image classification"**, ***MVA*, 2017**
5. Ali Diba, Ali Mohammad Pazandeh, Hamed Pirsiavash, Luc Van Gool, **"DeepCAMP: Deep Convolutional Action & Attribute Mid-Level Patterns"**, ***CVPR*, 2016**
4. Ali Diba, Ali Mohammad Pazandeh, Luc Van Gool, **"Efficient Two-Stream Motion and Appearance 3D CNNs for Video Classification"**, ***ECCV*, 2016**
3. Amir Ghodrati, Ali Diba, Marco Pedersoli, Tinne Tuytelaars, Luc Van Gool, **"DeepProposals: Hunting Objects and Actions by Cascading Deep Convolutional Layers"**, ***IJCV*, 2016**

2. Ali Diba, Amir Ghodrati, Marco Pedersoli, Tinne Tuytelaars, Luc Van Gool, **"Deepproposal: Hunting objects by cascading deep convolutional layers"**, ***ICCV*, 2015**
1. Mohammad Rastegari, Ali Diba, Devi Parikh, Ali Farhadi, **"Multi-Attribute Queries: To Merge or Not to Merge?"**, ***CVPR*, 2013**

Achievements

Awards/Recognition

- Outstanding reviewer for ICCV 2019, CVPR 2020,2021
- KU Leuven Ph.D. scholarship 2014-2020

Invited Talks

- ECCV workshop on Brave idea on video understanding, "Efficient Two-Stream Motion and Appearance 3D CNNs for Video Classification", Amsterdam, 2016
- Machine vision applications, "Deep Visual Words", Nagoya, Japan, 2017
- International Conference on Computer Vision, "Dynamic Action and Motion Network", Seoul, South Korea, 2019
- ICCV HVU workshop, "Holistic Video Understanding", Seoul, South Korea, 2019

Professional Activities

- Journal reviewer for IEEE TPAMI, IEEE TIP, IEEE TMM, IJCV, CVIU 2016-present
- Conference reviewer for CVPR, ICCV, ECCV, ICLR, BMVC, NeurIPS 2015-present
- Organizing member and chair of "Large-Scale Holistic Video Understanding (LSHVU) Workshop/Tutorials" at CVPR, ECCV, and ICCV venues.

Grants

- 2.9M€ (2022-2024), VLAIO research grant HBC.2022.0474, Research and development on computer vision solutions for cashier-less payment solutions.
- 110K€ (2021 - 2022), OCRE Cloud computation grant, to pursue research on self-supervised neural network.
- 440K€ (2020 - 2022), VLAIO R&D grant HBC.2020.2585, Research and development on on-edge computer vision solutions like video and scene understanding.
- 80K€ (2019 - 2020), KU Leuven internal research grant, creating a large-scale holistic video understanding dataset.
- 60K€ (2018 - 2019), EIC accelerator grant 835681.
- 120K€ (2017 - 2018), AWS Cloud computation grant, To facilitate research and development on deep learning solutions in visual understanding at scale.

Teaching Experinces

- *Image interpretation and pattern recognition, KU Leuven* *2015-2017*
  Senior teaching assistant in charge of designing exercises and organizing hands-on sessions.
- *Machine learning, Video encoding-decoding* *2013-2014*
  Teaching assistant at Sharif University

Skills

- *Programming languages*
  Python, C, C++, MatLab
- *Computer Vision and Machine Learning*
  MLOps, Pytorch, Tensorflow, Keras, CUDA, TensorRT, ONNX, OpenCV
- *Tools*
  Docker, AWSEC2, S3, Git, Jira, DataBricks, GCP, Azure
- *Soft Skills*
  Technical Leadership, Talent/ Project Management, Multi-functional Collaboration, Analytical/ Critical Thinking, Adaptability

References

- Prof. Luc Van Gool, ETHz, KU Leuven
  vangool@vision.ee.ethz.ch
- Prof. Rainer Stiefelhagen, Karlsruhe Institute of Technology
  rainer.stiefelhagen@kit.edu
- Prof. Ehsan Adeli, Stanford
  eadeli@stanford.edu
- Prof. Juergen Gall, Uni-Bonn
  gall@iai.uni-bonn.de